

Professional README - Polymer Property Prediction

Overview

This project was developed for the ANRF AISEHack 2.0 – Polymer Property Prediction Kaggle Competition. It predicts Glass Transition Temperature (Tg) and Chain Band Gap (Egc) directly from polymer SMILES strings using machine learning and cheminformatics.

Problem Statement

Build a machine learning regression model that predicts Tg and Egc from SMILES representations, reducing the need for expensive and time-consuming laboratory experiments.

Objectives

- Predict Tg accurately.
- Predict Egc accurately.
- Build a complete ML pipeline.
- Generate a Kaggle-compatible submission file.

Dataset

Training: 6171 samples (smiles, target, target_type).

Test: 4115 samples (id, smiles, target_type).

Workflow

SMILES → RDKit Feature Extraction → Molecular Descriptors → Feature Engineering → Machine Learning Model → Prediction → submission.csv

Technology Stack

Language: Python

Libraries: Pandas, NumPy, RDKit, Scikit-learn, Matplotlib

Environment: Kaggle Notebook, Jupyter Notebook

Machine Learning Pipeline

1. Load datasets.
2. Preprocess SMILES.
3. Generate RDKit descriptors.
4. Train regression model.
5. Evaluate using Mean R².
6. Predict test data.
7. Create submission.csv.

Evaluation Metric

The competition score is the average R² across Tg and Egc. Higher values indicate better predictive performance.

Applications

Sustainable polymer research, biomedical materials, flexible electronics, batteries, aerospace, automotive, and eco-friendly packaging.

Repository Structure

README.md
requirements.txt
.gitignore
LICENSE
notebook.ipynb
Polymer_Property_Prediction_Detailed_Report.pdf

Installation

```
git clone https://github.com/srimanreddy06/Polymer-Property-Prediction
cd Polymer-Property-Prediction
pip install -r requirements.txt
jupyter notebook
```

Results

Successfully trained a regression model, generated predictions for Tg and Egc, and produced a Kaggle-compatible submission.csv.

Future Improvements

LightGBM, CatBoost, XGBoost, ChemBERTa, Graph Neural Networks (GNNs), ensemble learning, and hyperparameter optimization.

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